Lab 9: Report 2 Peer Review Workshop

Reading each other's drafts of Report 2, and a plan for revising your own

FALL 2026 · WEEK 12 — WEEK OF MONDAY 2026-11-09, IN SECTION

DELIVERABLES — DUE AT THE END OF YOUR LAB SECTION, ON BRIGHTSPACE

- ☐ `Lab-9_FIRSTNAME_LASTNAME.Rmd/.qmd`: the starter sheet, renamed, with the three parts completed as far as you got and the generative AI disclosure filled in
- ☐ `Lab-9_FIRSTNAME_LASTNAME.pdf`: the same sheet, knitted to PDF
- ☐ any data or image file the sheet reads, if you add one

Report 2 (linear regression) is due on Monday 2026-11-16. Labs 6 to 8 drafted its parts: the outcome, the predictors and the exploratory figures in Lab 6; the question, the model's data, the first model, its checks and a second fit in Lab 7; and the final model's table, its sentences, its test and a first version of the answer in Lab 8. In this lab you show your draft to two or three classmates, read theirs, and leave with a list of changes to make before the deadline.

A regression report is a sequence of steps: a question about one predictor with the others held fixed, the hypotheses, a first model, its checks, the changes the checks led to, and a final model whose numbers answer the question. Each step depends on the one before it. An author who knows the analysis well can read past a missing step without noticing it. A classmate who reads the draft for the first time is more likely to notice the gap, so the questions in Part 2 ask, first of all, whether the steps connect.

The lab has three parts, as Lab 5 did. First, look back with your group at what you did with the comments on Report 1, and practice on an invented passage from a regression report, so that the group agrees on what a useful comment looks like before the drafts; second, take turns showing your drafts, and give each draft two stars and a wish; third, decide what to change in your own report and start on the first change.

About ten minutes to read, and the rest of the section with your group.

What to download and where to put it

Two files are on Brightspace for this lab:

- this handout;
- the starter sheet, `lab-09-report-2-peer-review-workshop.Rmd/.qmd`.

Please make a folder for the lab inside the course folder (`labs/lab-09/` for instance) and put the `.Rmd/.qmd` in it. The sheet is a written record of a conversation, so it reads

no data file. (If you add a figure or code that reads a file, please submit that file too, as in the earlier labs.)

Please bring your Report 2 draft as it is now, and have it open on your laptop with its `.Rmd/.qmd` and data. If it knits, please have the PDF open as well, since the PDF is what a reader of the report sees. Please also have the Analysis Report 2 brief open (it is on Brightspace, with its rubric), since the questions in Part 2 follow its sections. If the report has not been put together yet, the work from Labs 6 to 8 is enough to review: the choice of outcome and predictors and the exploratory figures from Lab 6, the question, the first model and the checks from Lab 7, and the table and sentences for your final model from Lab 8, Part 4. A question, its hypotheses and a first model are enough to start a useful conversation. You do not need slides or a prepared presentation.

What to hand in and when

Rename the starter to `Lab-9_FIRSTNAME_LASTNAME.Rmd/.qmd`, fill it in as your group works, knit it to PDF, and submit both files on Brightspace before you leave. Each person submits their own sheet: the comments you gave, the comments you received, and your plan. You submit your revised report separately, as Report 2. Its deadline is the following Monday, 2026-11-16, at 11:59 pm.

> **NOTE — How this lab is graded**
> The syllabus rubric applies, since this lab runs in section. Labs are out of 5 points: one for arriving before the section starts, three for progress (at least half of the sheet by the end of the section), and one for the TA checking your progress when you hand it in. Half of this worksheet is Part 1 and, from Part 2, the comments you gave one classmate and the comments you received on your own draft. Your draft is not graded in this lab. The comments you give are not a grade for anyone's report.

> **NOTE — If you miss your section**
> A peer review needs a classmate, so please ask your TA as soon as you can to arrange an exchange with one. (Under the syllabus, attending another lab section is not an option.) The exchange can be in person, on a call, or in writing. Then complete the sheet within one week of your original lab time for the three progress points; as with any missed lab, the two attendance points are not available. One review in each direction is enough: the comments you gave one classmate, and the comments they gave you. Please record the exchange that actually happened. The Report 2 deadline does not move, so an exchange before Monday 2026-11-16 will be the most useful to you.

> **NOTE — Your own comments**
> The syllabus does not allow generative AI for peer review, because the comments are meant to show your own reading of the draft. Please read each draft yourself and write your comments in your own words. Work on your own report is covered by the

> usual policy: generative AI is allowed there, with a disclosure at the end of the report. The disclosure at the end of this sheet covers this lab only.

> **HINT — If knitting to PDF will not work**
> The syllabus's fallback applies: knit to HTML, convert the HTML to a PDF with an online converter such as `https://www.sejda.com/html-to-pdf`, and submit that. Please also tell your TA, so that the LaTeX install is sorted out before Report 2 is due.

How the two hours are divided

The reviews take an hour. Everyone in the group has an equal turn in it.

Activity	Minutes
Read this handout	10
Part 1: a look back, and a practice review	15
Part 2: the drafts, one author at a time	60
Part 3: reflection, on your own and with your group	20
Part 3: the first revision	10
Knit, show your TA, and submit	5

Part 1: Feedback on a regression report

Before an analysis reaches a client, a supervisor, or the public, it is usually read by a colleague first. For a regression, that colleague asks questions such as *What is held fixed in that number? Is that the model you kept, or the first one? Did the answer change when you left those observations out?* Each of them asks whether a step of the analysis is on the page. This lab sets aside an hour for those questions, on a draft that is due on Monday 2026-11-16.

What makes a comment useful

As in Lab 5, a comment is useful when it names something specific in the draft and says why it matters to a reader. "Looks good" does not tell the author what to keep. "I don't like it" does not tell the author what to change.

A regression report invites one more kind of comment that is hard to use: a comment on the size of a number. A low R^2, or a p value above 0.05, is a finding about the data rather than a fault in the report. The brief gives no points for a high R^2 . An assumption that is not met costs no points, provided the report says so and says what follows from it. A comment on a number is useful when it asks whether the report reads the number correctly.

The examples below are about an invented draft. It asks how monthly rent, in dollars, is related to distance from campus, in miles, with floor area held fixed.

> **WARNING — Too general to use**
> "Your model looks good."
> "Maybe add more variables?"
> "Your R^2 is pretty low."

None of these says which section or number is meant, what works or does not work about it, or what to try instead. The second could be said about any regression. The third is about the data rather than the report.

> **HINT — Specific enough to use**
> "Section 5 points out that the spread of rent widens with floor area. Section 7 finds the same funnel in the residuals, so I could see why the robust standard errors are in Section 8 before I got there. Please keep that."
> "Section 9 says each additional mile from campus is associated with $48 less rent, but it does not say that floor area is held fixed. Without that, I read it as the slope in the scatterplot in Section 5. Could the sentence name what is held fixed?"
> "Section 7 gives the Breusch-Pagan test as p = 0.003 and says equal variance is met. I think the test's null hypothesis is that the spread is equal, so a small p value would be evidence that it is not. Could you check that judgment?"
> "Section 8 says three apartments were left out, but not which ones or why. Could you give distance's slope with them and without them? Then I could tell whether the answer depends on them."

Each of these names a section and says why the feature matters to a reader. The first explains why something works, so the author knows to keep it. The second and third name what a reader would misread. The third is phrased as a question, because a reviewer can be wrong about a test too. The fourth asks for the number that would let a reader judge a choice the author made.

Two stars and a wish

For each draft in Part 2, please give **two stars and a wish**:

- **Two stars**: two specific things that work in the draft, each with the reason it works. "Your Section 8 gives distance's slope before and after each refit, so I could see that the answer did not depend on the three apartments" is a star. "It's good" is not.
- **One wish**: one specific change the author could make, phrased as a suggestion or a question rather than a verdict. "Have you considered…", "I wonder what would happen if…" and "Could you…" are good ways to begin. A wish names a section, and a change the author could make there before the deadline. A change that needs a new dataset, or a model of a different kind, cannot be made by then, so please offer it as a future direction for the Discussion instead.

The stars come first, because they tell the author what to keep. An author who hears only what to change can end up changing the parts of the report that were already working.

> **NOTE — The author decides**
> You do not have to act on every wish you receive. Part of receiving feedback is hearing a suggestion out, considering it, and then deciding what to do with it. Some wishes point to a real gap in the report. Others reflect a difference in taste. Part 3 asks you to record any suggestion you decide against, and why.

Ground rules

Showing work that is not finished can feel uncomfortable. Everyone in your group is in the same position, since every report is still a draft. The aim of the session is for each author to leave with ideas for the next version. The five ground rules from Lab 5 still apply:

1. **Be specific.** Name the section, the number, or the sentence, and say why it works or what is unclear about it.
2. **Suggest rather than prescribe.** "Have you considered…" opens a conversation about a choice. "You should…" makes the choice for the author.
3. **Stars first**, even when the wish seems more urgent.
4. **Listen before responding.** When you are the author, please let each reviewer finish, and take notes. It is tempting to explain or defend the draft straight away, but please keep your questions until the comments are done.
5. **Respect the work.** Each draft represents hours of its author's thought and effort.

> **TASK 1 — A look back, and a practice review (15 minutes)**
> Please form a group of three or four (a group of five is fine if the numbers do not divide evenly). Please include at least one person who did not read your Report 1 draft in Lab 5, so that someone in the group comes to your analysis new. Take turns answering these two questions aloud, then write your own answers in Part 1 of the sheet.
>
> 1. Which comment on your Report 1 (in Lab 5, or with the marked report) did you act on, and what did you change? If you decided against one, what was your reason, looking back?
> 2. What would you like this group to know about how you prefer to receive feedback? It is fine to say what you said in Lab 5.
>
> Then please read the invented passage below on your own. It has more than one problem, so a wish means choosing the change that would help a reader most. Please

write two stars and a wish for it in the sheet, then compare them with your group. Did you choose the same wish, and why?

NOTE — An invented passage, from Sections 9 and 10

9. The final model. We kept the model with robust standard errors (Section 8). It accounts for 41% of the variance in monthly rent ($R^2 = 0.41$, adjusted $R^2 = 0.40$). Its residual standard error is \$310, so a typical apartment's rent is about \$310 from its prediction. Each additional mile from campus decreases rent by \$48 ($p = 0.002$). Because the p value is below 0.05, distance has a large effect on rent, and we reject H_0.

10. Discussion. Landlords charge less for apartments farther from campus, so students who want to save money should live farther away. One limitation is that there could be confounding variables.

Knowing how each person in your group prefers to hear a comment helps you phrase yours for them in Part 2.

Part 2: Reviewing the drafts

Part 2 takes most of the section. Each person in turn shows their draft. The rest of the group comments on it with two stars and a wish. Please review the draft as it is now. You do not need to predict how the finished report will turn out, or to give it a mark. You do not need to prepare a presentation either. Open the report on your laptop and talk the group through what you have, as you would show a colleague something you have been working on.

What to look for

The Analysis Report 2 brief lists what the report contains, in eleven sections. The questions below follow it. You do not need to ask all of them about every draft. Please start with three places that have to agree: the question and H_0 in Section 2, the predictor of interest's sentence and its test in Section 9, and the summary at the start of the Discussion. Do all three name the same outcome, the same predictor of interest, and the same predictors held fixed? Is the number in the Discussion the one in the final model's table? Then please choose the two or three places where a comment would help the author most.

- **The question and the hypotheses (Section 2).** Does the question end in a question mark, name the outcome and the predictor of interest in their units, and name the predictors held fixed? Does it ask about a relationship rather than a cause? Is H_0 about the predictor of interest's slope, with the other predictors in the model? Does H_A give the direction expected, with a reason (Lab 7, Part 1)?
- **The data the model uses (Sections 1, 3 and 4).** If the dataset is the one from Report

1, has the card been updated for the rows this model uses? Is every variable in the model described, and only those? Is a categorical predictor stored as a factor, with its reference group named? Is there one table of the rows observed on every variable in the model, with its row count, so that every model is fitted to the same rows (Lab 7, Part 1)?

- **The exploratory analysis (Section 5).** Does the paragraph say what the figures show for the model: a curve, a spread that changes, values to check, or two predictors that are closely related? Please read it again after Section 7. Did the checks find anything that the figures could have shown first (Lab 6, Parts 2 & 3)?

- **The first model (Section 6).** Is there a reason for each predictor? Are R^2, adjusted R^2, and the F test with its degrees of freedom and p value reported (Lab 7, Part 2)? The coefficients are interpreted for the final model, so a first model with no sentences about its coefficients is what the brief asks for.

- **The checks (Section 7).** Are the checks made on the residuals rather than on the outcome? Does each test have its statistic, its p value, and its null hypothesis? Each test's null hypothesis is that its assumption holds, so is a small p value read as evidence that the assumption is not met? Was the Durbin-Watson test run on rows in an order that means something, or does the report say the rows have none (Lab 7, Part 3)? Are the variance inflation factors there? For influence, does the report give the number of observations past each rule of thumb, and name the ones past all three (Lab 7, Part 4)? Does each judgment say what a failure costs?

- **Remediation (Section 8).** Does each check that failed have a remedy, or a reason for none? After each change, does the report compare the predictor of interest's coefficient (its sign, its size and its p value) and R^2 before and after? Is the comparison without the observations past all three influence rules there, even if every check passed? If observations were left out, can you tell which ones, how many out of how many, and why (Lab 7, Part 4; Lab 8, Part 4)?

- **The final model (Section 9).** Does each coefficient's sentence have the four pieces from Lab 8, Part 1: the step in the predictor with its units, the difference in the outcome with its direction and units, what is held fixed, and a verb that describes a comparison rather than a cause? Is each step one a reader can picture, or would the predictor read better rescaled (Lab 8, Part 1)? Does the report say what the predictor of interest's p value is the chance of, and give ΔR^2 and the F test from `anova()`? If the model has an interaction, a logarithm or a curve, is it read as in Lab 8, Parts 2 and 3? Does the section end with the state of the four assumptions for this model?

- **The Discussion (Section 10).** Does the summary answer the question with the final model's numbers, in words a reader without statistics could follow? Do the implications claim no more than an association in observed data? Do the limitations name a variable that is not in the model and could explain part of the

slope, rather than "there could be confounding variables"? Is the future direction specific enough that someone else could carry it out (Lab 4, Part 2)?

- **Code and output.** Is the code for the refits hidden with `echo = FALSE`, with the text saying what each step found? Is anything printed that a reader does not need, such as pages of `summary()` output or the dataset (Lab 1, Part 2)? If a number in the text does not match the table, please ask the author to show you the line that computes it.

Report 2 asks for the checks and a judgment on each, not for data that pass them. A failed check, and a remedy that did not work, are both fine when the report says what happened (the brief's section on Remediation names three such cases). So please comment on whether the report says what happened, rather than on whether the checks passed.

A model with additive predictors is enough for Report 2. In Lab 8 we treated interactions, logarithms and curves as optional, because they make the interpretation harder, and advised against splines for this report. So please do not suggest one for its own sake.

Each author's turn

The hour is divided equally among the group: 20 minutes for each author in a group of three, 15 in a group of four, and 12 in a group of five. The times in the task below are for a group of four. Please lengthen or shorten each step to fit your group, and keep to the turns so that the last author has as much time as the first.

TASK 2 — One author's turn (15 minutes in a group of four)

First, the author gives the answer and shows the draft (4 minutes). Before your turn, please fill in "Your draft" in Part 2 of the sheet. Then please start with the answer, in a minute or less: the question, and what your final model says about the predictor of interest, in its units and with what is held fixed. Then show the group the final model's table, what the checks found, and what you changed in Section 8. Please say why you made each choice, because the group can only comment on a choice if they know the reason for it. End with the one thing you would most like help with. While the author talks, everyone else listens and takes notes.

Second, say it back and follow one number (2 minutes). One reviewer says the predictor of interest's sentence back in their own words, as a comparison of two observations. For the rent draft: "Take two apartments with the same floor area, one a mile farther from campus than the other. The model predicts about $48 a month less for the farther one." If the reviewer cannot say it that way, the sentence is the first place for a wish. Another reviewer finds the predictor of interest's coefficient in the Discussion and in the final model's table. Are they the same number, from the same model? Each reviewer records what they found in the sheet.

Third, the stars (2 minutes). Each reviewer says one of their two stars aloud: something that works, and why. If someone has already named your star, add your reason to theirs or give your other one. Each reviewer records both stars in the sheet.

Fourth, the wishes (4 minutes). Each reviewer offers their wish. The group then talks over which one or two changes would help the report most. The author listens first, then asks about anything that is unclear. Each reviewer records their wish. The author records the stars and wishes they received.

Fifth, a variable not in the model (3 minutes). As a group, please name one variable that could explain part of the relationship between the predictor of interest and the outcome, because it is related to both (Lab 6, Part 3). It may be in the dataset or not. Would the predictor of interest's coefficient be larger or smaller with that variable held fixed as well, and why? If the variable is in the data, the author could add it in a sensitivity comparison, fitted to the same rows. If it is not, it is a limitation specific to this report, for the Discussion. This step is about what the data cannot rule out, not about what is wrong with the model.
Then move on to the next author.

For the rent draft, the fifth step might go like this: "Apartments farther from campus might be in newer buildings, which rent for more. If so, newer buildings make up part of the rent the farther apartments would otherwise lose. With the building's age held fixed, distance's slope would be further below zero than −$48."

Lab 8, Part 1: a coefficient is the difference between the predicted outcomes of two observations that differ by one unit in one predictor and have the same values of the others. A sentence that can be said back in that form has its four pieces.

NOTE — If a draft is not finished

A question with its hypotheses, a first model, or the checks from Lab 7 can each be reviewed. Please say what the work so far makes clear, and suggest the next step to take. If there is no final model yet, please say back the sentence from the author's Lab 8 sheet, or the first model's coefficient for the predictor of interest, in the second step. Where a section is missing, please say that it is missing, so that the author can tell a comment on the draft from an idea for work they have not started.

HINT — If your group finishes early

Please go back to the section that drew the most discussion and work on it together. Write the predictor of interest's sentence again, or the paragraph for Section 8, and compare it with the original.

Part 3: Reflection and revision

While the conversation is fresh, please take fifteen minutes on your own to write down what you heard and what you will do about it, then five minutes with your group on how the feedback was given. The last ten minutes are for starting the first change.

TASK 3 — On your own (15 minutes)

Part 3 of the sheet asks three things:

1. **Which comments mattered most?** The one or two comments that made you think "I need to change that" or "I hadn't considered that", and why.
2. **What will you change?** Two or three specific changes, in the order you will make them, each with the section it is in and how you will check that it worked. "Fix my interpretation" is not specific. "Add 'with floor area held fixed' to distance's sentence in Section 9 and change 'decreases' to 'is associated with', then ask someone who has not seen the report to say the sentence back" is specific. If you decided against a suggestion, please say why. For example, adding a variable the group suggested might leave fewer rows observed on every variable, so every model would have to be fitted again. Naming it as a limitation may be the better change before the deadline.
3. **What did you learn from someone else's draft?** Not from the comments you received, but from seeing how someone else approached a similar problem: a way of writing up Section 8, for instance, or a question framed so that its answer was easy to find in the Discussion.

TASK 4 — With your group (5 minutes)

Please talk over these three questions, then write your own answers in the sheet. It is fine to say that one of them did not come up in your group.

1. Did giving feedback change how you see your own report? For example, sometimes saying someone else's coefficient back shows you that your own sentence is missing a piece.
2. Did the way someone phrased a comment (to you, or to someone else in the group) make it easier to use?
3. Was a comment on the statistics (a test, a check or a remedy) harder to give than a comment on the writing? If it was, what helped?

Giving clear, specific, constructive feedback is not something that matters only in school. *It is one of the most transferable professional skills you can develop.* Almost every job involves giving and receiving feedback, and the people who do it well are the people others want to work with.

The first revision

Please use the last ten minutes of the section to make the first change on your plan, in your report, while the comments are fresh. If the change is to a number in the text, please make it with inline R, so that the number changes when the model does. In the sheet, record what you changed, how you checked it, and what remains to do before the deadline. If the change will not be finished today, record the first step you

took and what you still need.

When you have finished

Knit the sheet and check two things in the PDF: each comment you recorded names its section, and each change in your revision plan has a check beside it. Then show your TA where you got to, and upload the `.Rmd/.qmd` and the PDF to Brightspace before you leave.

> **READING — Reference**
>
> - The Analysis Report 2 brief and its rubric, on Brightspace, for what the report contains.
> - Lab 6, Parts 2 and 3, for the exploratory analysis; Lab 7, for the question, the first model and its checks; Lab 8, for the sentences and the final model; Lab 4, Part 2, for the Discussion; Lab 1, Part 2, for the sections of a report and what not to print.

SUBMISSION

- ☐ Knit your document to PDF
- ☐ Check that all code is visible (not running off the page)
- ☐ Check that all output displays correctly
- ☐ Submit both `.Rmd/.qmd` and `.pdf` files